

LENOUVEL LOUIS

Final-year Engineering student at ISEP, specializing in Artificial Intelligence

+33 6 65 44 88 94 | mr.lenouvel.louis@gmail.com | [Louis Lenouvel](#) | [GitHub](#) | [Portfolio](#)



Rigorous, autonomous and results-driven, I am completing 3 years of professional experience in Data Engineering. I am now moving toward an Artificial Intelligence role from September 2026

PROFESSIONAL EXPERIENCE

Data Engineer, IKIGAI Games for Citizens

Paris, France | Sept. 2023 - Sept. 2026 (3 years)

- Design of an xAPI pipeline processing over 5 million educational usage records, with generation of ready-to-use dashboards
- 20% reduction in data errors through automated quality checks and data standardization
- Production deployment of a fully local RAG system (Ollama/Mistral, ChromaDB) over PDF, DOCX, PPTX and HTML: Tesseract OCR, BM25+RRF hybrid search, cross-encoder reranking, HyDE, multi-hop
- Reliability and industrialization: anti-hallucination safeguard, sentence-level citation traceability, golden evaluation with a blocking regression gate in CI, observability and quality-drift alerting
- Setup of dedicated GPU infrastructure (VM, PCI passthrough VFIO/IOMMU) and resolution of an NVIDIA driver issue, cutting indexing time from ~1 min to 7.3 s

EDUCATION

Engineering Degree (MSc level), ISEP

Paris, France | 2023 - 2026

Specialization in Artificial Intelligence: Computer Vision, Machine Learning, Deep Learning, Optimization

Academic exchange semester, Hanze University of Applied Sciences Groningen, Netherlands | Feb. - Apr. 2026

Track: System Modeling, Leadership & Intercultural Teamwork

BSc in Computer Science, Université Paris Cité (Descartes)

Paris, France | 2021 - 2023

Development & Data track

PROJECTS

DeepRetrieval, RAG pipeline from scratch (Solo)

3 months

- Development of a complete RAG pipeline with no framework (ChromaDB, sentence-transformers, BM25+RRF hybrid search, re-ranking)
- MLflow evaluation (Recall@k, MRR, LLM-as-judge) on 45 Q/A pairs, embeddings x chunking experiments
- Production setup: FastAPI API, Docker, CI/CD GitHub Actions, Streamlit UI

MetroVision MLOps, ML production pipeline (Team)

4 months

- Industrialization of a vision system (CNN + k-NN) into a deployable, observable, reproducible service
- Kubernetes deployment (minikube + OpenShift) with Prometheus/Grafana monitoring
- Drift detection with Evidently AI, MLflow tracking, CI/CD GitHub Actions

PowerShift, Energy grid simulation platform (Team)

Hanze University, Netherlands | 3 months

- Async FastAPI/SQLAlchemy backend, PyPSA modeling, Azure deployment (Key Vault, NeonDB)

SKILLS

Languages: Python (Pandas, NumPy, scikit-learn, TensorFlow, PyTorch)

Machine Learning: supervised/unsupervised models, evaluation & metrics, feature engineering

Deep Learning & Vision: CNN (EfficientNet), OpenCV

GenAI / RAG: ChromaDB, sentence-transformers, Ollama, hybrid search (BM25+RRF), reranking cross-encoder, HyDE, OCR (Tesseract), LLM-as-judge

Data Engineering & BI: SQL (PostgreSQL, MySQL), async SQLAlchemy, FastAPI, ETL, xAPI ingestion, Clean Architecture, Power BI (DAX)

MLOps & Deployment: Docker, docker-compose, Kubernetes, OpenShift, GitHub Actions, GitLab CI, MLflow, Hugging Face, Prometheus, Grafana, Evidently AI, Azure, KVM/libvirt (GPU passthrough VFIO), systemd, Ansible

Other: Angular, TypeScript, Node.js, APIs REST, Git/GitHub/GitLab, pytest, VS Code, Jupyter, Gradio, Streamlit

LANGUAGES

French: native | English: B2 (TOEIC 885) | German: basics

INTERESTS

- Judo (6 years), training for black belt | Travel and Erasmus (Germany, UK, Netherlands) | AI technology watch